Quantifying the defects in the legacy PUV processing code

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Abstract

The legacy PUV processing code (`beachpeeps/PUV_Processing`) contains five identified faults. This note measures what each one does to the derived wave parameters, so that results produced with that code can be judged individually rather than dismissed or trusted wholesale. Each legacy choice was implemented as an isolated switch applied to identical input, making every difference attributable to one fault. Across 3,713 hourly segments from ten instrument records (record-median depths 5.5–11.7 m, four sites, segment H_s up to 4.6 m): significant wave height and mean wave direction are usable, with stateable offsets; peak period differs by definition rather than by error and is the largest single discrepancy between the pipelines; and the directional second moments, from which radiation stress and alongshore forcing are built, are the products that should not be relied on. Two results contradicted the expectations formed from reading the code, and one diagnostic in an earlier version of this comparison overstated a fault by a factor of three.

Who this is for. Anyone deciding whether a number produced by the legacy code (`beachpeeps/PUV_Processing`, used for earlier lab work including Ruby2D) can be reused. The summary table in Section 7 is the short answer. For the current pipeline, read `README.md` and `docs/pipeline_levels.md`; the code reading that first identified the faults is `docs/pipeline_comparison_legacy.md`.

Status, October 2026. The numbers were computed on 2026-07-27 from the L1 and L2 files of that date and were not rerun on catalog 2026-10-14.1. Each statistic is a difference between two estimators applied to the same input segment, so the later input corrections (headings, `doffp`, clocks, the edge-gap fill) are expected to move them little, but that has not been checked. All tabulated values were reproduced on 2026-10-15 from the saved result file (`outputs/validation/legacy_defect_isolation.mat`). Three statements in the July text did not match it and are corrected below: the number of sites (four, not five), the largest sampled H_s (4.59 m, from a record whose maximum is 4.66 to 4.67 m), and the claim that no valid segment had a pressure gap above 1 % (five did). One record, RUBY22/MOP579_6m, is now excluded from reported catalog statistics (`shared/excluded_records.m`); it contributes 166 of the 3,713 segments here.

1 Why isolate rather than compare pipelines

Running both pipelines end to end and differencing the outputs confounds all five faults together with several deliberate design differences. The resulting scatter shows *that* the pipelines disagree but cannot attribute the disagreement, and the attribution is the practical question: a result built on H_s may be sound while one built on radiation stress is not.

Each legacy behavior was therefore reimplemented from the legacy source as a switch applied to the same input segments, with everything else held at the current pipeline's settings. The faults follow the numbering in `docs/pipeline_comparison_legacy.md`.

D1: auto- and cross-spectra estimated differently. Legacy auto-spectra (`calculate_fft2`) use a Hanning window with 50 % overlap (`num` $= \lfloor 2n/\texttt{nfft} \rfloor - 1$ ensembles). Legacy cross-spectra (`cospec`) use a Hamming window with *zero* overlap and a hardcoded `nfft` = 2400 carrying the source comment *"changed from 7200. Why?"*. The two therefore differ in spectral window *and* in degrees of freedom, while

$$a_1 = \frac{\operatorname{Re} S_{pu}}{\sqrt{S_{pp}\,(S_{uu} + S_{vv})}} \tag{1}$$

divides one by the other. Every directional quantity inherits this.

D3: NaN replaced by zero before the FFT. `calculate_fft2.m:11` and `cospec.m:22`. The expectation recorded previously was an H_s deficit proportional to gap fraction.

D4: pressure correction. Legacy caps K_p^2 at 10; the current pipeline zeros frequencies where $K_p < 0.1$.

Design differences. The swell band runs to 0.20 Hz rather than 0.25 Hz, and peak period is an energy-weighted centroid rather than the frequency of the spectral maximum. Neither is a fault. In this comparison both period definitions are evaluated on the same multitaper pressure spectrum.

Not quantified. D2 (frequency-grid mismatch between `fm` and `f_co`) and D5 (`Suv` indexed at line 49, computed at line 107) are structural faults with no correct magnitude to measure. D5 either errors or silently reuses a stale workspace variable, so its effect depends on execution history rather than on the data.

2 Data

Ten instrument-records, 3,713 hourly segments: up to 400 valid segments per record, evenly spaced through the record, with segments more than half NaN skipped. The records are at four sites: Torrey Pines (TBR23, TOR19W, TOR20W, TOR23W, RUBY22 and the offshore MOP591 station, TOR14B and TOR15B), Imperial Beach (IB18W), Coronado (COR16B) and SIO Pier (SIO25E). Two records (TOR19W and TOR20W, MOP582, 10–11 m, 2019–2021) match the station, depth and period of published work using this code, so the deltas there transfer directly. Five records were added specifically to reach conditions the first pass could not: large waves (TOR15B, record maximum H_s 4.67 m in the July L2 and 4.66 m in catalog 2026-10-14.1; COR16B, 4.19 m, which also has 24 % of its velocity missing), and whole-record pressure-gap fractions of 11 %, 35 % and 85 % (TOR14B, TOR23W/MOP586_10m, RUBY22/MOP579_6m).

3 D1: the estimator mismatch

Mean direction is essentially unbiased in the median because the fault injects variance rather than a systematic offset, though individual hours reach 6.6°. The second moment does not fare as well: b_2 has a 95th-percentile error of 0.326 on a quantity that spans only $[-1, 1]$. Radiation stress

$$S_{xy} = \rho g \int S(f)\,\frac{c_g}{c}\,\frac{b_2(f)}{2}\,df \tag{2}$$

is built directly from b_2 and inherits that error in full, which makes legacy alongshore forcing the least reliable product of that pipeline.

Table 1: Legacy minus current, directional quantities. a_1 and b_2 are evaluated at the spectral peak and are bounded on $[-1, 1]$.

Quantity	Median	IQR	95th pct. of $\lvert\Delta\rvert$
Mean direction D_p (deg)	−0.07	−1.40 to +1.40	6.6
Spread, peak bin (deg)	−4.32	−7.78 to +29.03	50.3
Spread, band-averaged (deg)	+0.26	−9.45 to +6.24	**17.1**
a_1 at peak	−0.001	−0.043 to +0.040	0.177
b_2 at peak	+0.002	−0.080 to +0.085	**0.326**

A correction to the earlier version of this comparison

The peak-bin spread figure above (95th percentile 50.3°) overstates the fault by roughly a factor of three, and an earlier draft of this comparison reported it without qualification. Spread computed at a single bin from

$$\sigma_1 = \sqrt{2\,(1 - r_1)}, \qquad r_1 = \sqrt{a_1^2 + b_1^2} \tag{3}$$

is dominated by the degrees-of-freedom difference between the two estimators rather than by the window mismatch: with zero-overlap cross-spectra the peak-bin r_1 is noisy and occasionally exceeds unity, where it must be clamped. Averaging the complex first moment over the swell band before taking its modulus,

$$\bar{r}_1 = \left| \frac{\sum_f S(f)\,[a_1(f) + i\,b_1(f)]}{\sum_f S(f)} \right|, \tag{4}$$

suppresses that noise and isolates the window mismatch. The band-averaged statistic is the defensible one and is closer to the form in general use following Kuik et al. (1988). The fault is real, and 17° at the 95th percentile is still large, but it is not 50°.

Dependence on conditions

Table 2: Median $\lvert\Delta\rvert$ by significant wave height.

H_s (m)	n	$\lvert\Delta D_p\rvert$ (deg)	$\lvert\Delta\sigma_1\rvert$ (deg)	$\lvert\Delta b_2\rvert$
< 0.5	374	1.71	6.22	0.129
0.5–1.0	1816	1.48	7.07	0.095
1.0–1.5	889	1.34	8.14	0.066
1.5–2.0	359	1.33	8.63	0.059
2.0–3.0	247	1.11	7.79	0.065
> 3.0	28	0.83	6.62	0.099

The direction error falls monotonically with wave height, from 1.71° below $H_s = 0.5$ m to 0.83° above 3 m, and $\lvert\Delta b_2\rvert$ falls by half over the same range. This is the signature expected of a variance-driven fault: at higher signal-to-noise the two estimators converge. **Large waves are the regime in which the legacy directional output is most trustworthy, not least.** The apparent uptick in $\lvert\Delta b_2\rvert$ in the highest bin rests on 28 segments and should not be read as a reversal.

The error grows with directional spread instead, from a median $|\Delta D_p|$ of 1.35° below 15° spread to 2.79° at 20–25°: broader seas have lower coherence per frequency, so the estimator difference matters more.

4 D3: the NaN handling, and why it does not bite

This fault could not be made to act as documented, and the reason is more useful than the number would have been.

Records were selected specifically for pressure gaps: 11 %, 35 % and 85 % NaN in the raw pressure. After the standard validity gate, 3,708 of the 3,713 sampled segments (99.9 %) have a pressure-gap fraction below 1 %. The segments in which zeroing would matter do not reach the spectral estimator, because the existing quality control rejects them first.

Where small gaps do survive, the fractional change in H_s is erratic rather than a deficit: the median is zero, the interquartile range reaches +0.19, and gap fraction barely predicts the change (Spearman $\rho = -0.06$, $p = 0.04$, $n = 1152$ segments with any pressure gap). The predicted mechanism, variance removed in proportion to the gap, competes with a second effect in the opposite direction, since zeroing a segment of an oscillating record introduces step discontinuities that inject broadband energy.

The gaps in these records are overwhelmingly in *velocity*, not pressure: 8.8 % NaN in U and V on TOR19W against 2×10^{-5} in pressure, and 24 % on COR16B. So the fault does not act on H_s at all. It zeros roughly one velocity sample in eleven before the FFT, which removes velocity variance and therefore acts on the directional coefficients and on U_b, compounding D1 on exactly the quantities D1 already damages. The original write-up attributed this fault to the wrong variable.

The protection here is the validity gate, not the NaN handling. Any reprocessing that relaxes that gate would expose this fault.

5 D4: the pressure-correction rule

Table 3: Median $|\Delta H_s|/H_s$ from capping K_p^2 at 10 rather than zeroing frequencies below $K_p = 0.1$.

Depth (m)	n	Median $\lvert\Delta H_s\rvert/H_s$
< 6	339	0.000
6–8	867	0.000
8–10	1387	0.005
> 10	1120	**0.015**

This runs opposite to the expectation recorded from the code reading, which argued from noise amplification being permitted in shallow water. Measured on H_s, the two rules diverge in *deeper* water: there K_p falls below the cutoff across a wider portion of the sea-swell band, so the capped and zeroed treatments disagree over more frequencies. The effect is 1.5 % above 10 m and negligible below 8 m.

The shallow-water concern in the original reading is not thereby dismissed. It was about noise admitted at high frequency, which affects the spectral tail and any quantity weighted toward it; H_s is an integral dominated by the peak and is a poor detector of it.

6 Design differences, which are larger than most of the faults

Swell-band upper limit. Truncating at 0.20 Hz rather than 0.25 Hz makes legacy H_s^{SS} 2.9 % low in the median, rising to 7.8 % for $T_p < 8$ s and falling to 2.5 % for $T_p > 16$ s. A wind-sea effect, negligible in swell.

Peak period. The energy-weighted centroid sits 2.73 s below the argmax definition in the median, with an interquartile range of −4.29 to −1.47 s and a 95th percentile of 7.11 s. **This is the largest single difference between the pipelines and it is not an error in either.** The two quantities are different statistics of the same spectrum. A legacy T_p compared against a model T_p, or against a value from another study, is a comparison of definitions unless the centroid is stated explicitly.

7 Summary

Quantity	Status	Qualification
H_s	Usable	2.9 % low from the band limit; a further 1.5 % above 10 m depth from the K_p rule. Both are offsets, not errors.
Mean direction D_p	Usable	Median bias below 0.1°; individual hours reach 6.6°. Best at large H_s, worst in broad seas.
T_p	Different, not wrong	2.73 s below an argmax T_p. State the definition before comparing with anything.
Directional spread	Distrust	17° at the 95th percentile, band-averaged.
b_2, S_{xy}, alongshore forcing	Do not use	95th-percentile b_2 error of 0.33 on a range of 2, with D3 compounding it through the velocity channel.

8 Limits of this comparison

Each fault was emulated from the legacy source rather than by executing that pipeline end to end, so interactions between faults would not appear here. Ten records at four sites is a narrow span: only 28 segments exceed $H_s = 3$ m, and only five sampled segments carry a pressure-gap fraction above 1 % (the largest 41 %), so the two regimes this note set out to probe are covered thinly at one end and not at all at the other. D2 and D5 remain unquantified.

The band-averaged spread reported here should be preferred over the peak-bin figure, including over the peak-bin figure that appeared in the earlier version of this comparison.

Script: `validation/archive/legacy_defect_isolation.m` (moved there on 2026-07-30; since 2026-10-15 it finds the repository root from either

folder). Result file: `outputs/validation/legacy_defect_isolation.mat` (local, not in the repository). Legacy source, outside this repository: `Research/Beach_Change_Observation/Vector/PUVs/PUV_Processing-main/Level2_QC/`. Prepared as internal documentation of why the legacy codes are deprecated.